

Supplemental Material: Subspace methods for min-max problems

Morteza Kimiaei

*Fakultät für Mathematik, Universität Wien
Oskar–Morgenstern–Platz 1, A–1090 Wien, Austria
email: morteza.kimiaei@univie.ac.at*

Shima Shabani

*Institute for Mathematics, Brandenburg University of Technology
Platz der Deutschen Einheit 1, 03046 Cottbus, Germany
email: shima.shabani@b-tu.de*

Michael Breuß

*Institute for Mathematics, Brandenburg University of Technology
Platz der Deutschen Einheit 1, 03046 Cottbus, Germany
email: breuss@b-tu.de*

September 15, 2026

Contents

1	Background on Monotone Equations, Min–Max Problems, and Learning Applications	3
1.1	Nonlinear monotone equations	3
1.2	Connection with min–max problems	4
1.3	Learning-based application classes	4
2	Asymptotic and Canonical OGDA-Type Direction Structures	6
2.1	Representative OGDA-type methods	9
2.2	Asymptotic OGDA-type structure of subspace directions	10
2.3	Unification of OGDA-Type Methods within the proposed PLS–S/PF–S Framework	15

2.3.1	Canonical asymptotic OGDA-type direction structure . .	15
2.3.2	Unified convergence-rate implications	17
3	Learning-Based Test Problems	18
4	A comparison between the JFS2dir variants	25
5	A comparison between the JFS3dir1 and JFS3dir2 variants	26
6	A comparison between the JFS3dir3 variants	27
7	Efficiency Improvements of Fixed-Step over Line-Search Variants	28
8	A comparison between OGDA-type variants	31
	References	32

The supplementary material provides additional background, structural analysis, implementation details, and numerical results that complement the main paper [9]. To make the presentation self-contained, we first recall the non-linear monotone equation problem studied in the main paper, its connection with convex-concave min-max formulations, and the three learning-based application classes used in the numerical experiments. We then examine the asymptotic and canonical OGDA-type direction structures underlying the proposed subspace framework. In particular, we review representative OGDA-type methods, establish an asymptotic residual-memory representation for the Jacobian-free subspace directions, and show how several classical OGDA-type schemes can be interpreted within the same canonical directional framework under appropriate additional conditions. Next, we give the complete construction of the learning-based test problems, including their dimensions, parameter ranges, known solutions, Lipschitz constants, initialization procedure, stopping criterion, and numerical validation. Finally, we report detailed comparisons among the two- and three-term Jacobian-free subspace directions, including both the line-search PLS-S and fixed-step PF-S variants, together with the comparative results for the OGDA-type methods. These supplementary results provide the structural details and detailed numerical

evidence supporting the theoretical and numerical summaries presented in the main paper.

1 Background on Monotone Equations, Min–Max Problems, and Learning Applications

For completeness and to make the supplementary material self-contained, we briefly recall the nonlinear monotone equation problem, its connection with convex–concave min–max problems, and the three learning-based applications used in the numerical experiments. These formulations and their role in the proposed framework are introduced in the main paper [9]; here, they are restated only to provide the notation and background needed for the supplementary numerical results.

1.1 Nonlinear monotone equations

As defined in the main paper, the basic problem considered throughout this work is the nonlinear monotone equation

$$E(x) = 0, \quad x \in \mathbb{R}^n, \quad (1)$$

where $E : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a monotone operator, i.e.,

$$\left(E(x_1) - E(x_2)\right)^\top (x_1 - x_2) \geq 0, \quad \forall x_1, x_2 \in \mathbb{R}^n. \quad (2)$$

The theory of monotone operators was originally developed in the 1960s in the context of functional analysis and partial differential equations. It was soon recognized that this framework is also fundamental for convex optimization. In the 1970s, iterative algorithms based on monotone operator theory—such as proximal-point and fixed-point methods—were introduced for solving convex minimization problems; see [15] and the references therein. Since then, the field has grown substantially and remains highly active, with applications in compressed sensing [6, 17], power engineering systems [16], and optimization models arising in machine learning [8].

The present work focuses specifically on the equation formulation (1). In this paper, the numerical models are constructed so that the relevant optimality systems are represented by the nonlinear monotone equation (1).

1.2 Connection with min–max problems

As discussed in the main paper [9], an important motivation for nonlinear monotone equations in modern machine learning is the transition from single-objective minimization to min–max and saddle-point formulations. Many learning and equilibrium problems can be expressed as

$$\min_{u \in \mathbb{R}^{n_u}} \max_{v \in \mathbb{R}^{n_v}} \Phi(u, v),$$

where $\Phi : \mathbb{R}^{n_u} \times \mathbb{R}^{n_v} \rightarrow \mathbb{R}$ is the corresponding objective function.

Under the standard differentiability and convex–concave assumptions on Φ , the associated first-order saddle operator is defined by

$$E(x) = \begin{bmatrix} \nabla_u \Phi(u, v) \\ -\nabla_v \Phi(u, v) \end{bmatrix},$$

where

$$x = (u, v) \in \mathbb{R}^n, \quad n = n_u + n_v.$$

The convexity of Φ with respect to u and its concavity with respect to v imply that this operator is monotone on $\mathbb{R}^n$ [2].

The first-order optimality conditions reduce to $E(x^*) = 0$, and hence have exactly the form of the nonlinear monotone equation (1). The equation setting considered in the main paper and in the supplementary numerical experiments therefore corresponds directly to unconstrained saddle-point systems represented by roots of the monotone operator.

1.3 Learning-based application classes

The three learning-based application classes used in the numerical experiments were introduced in the main paper. For completeness, we recall their min–max formulations here before describing the concrete test instances generated from them.

- **Multi-Agent Reinforcement Learning (MARL)** [7, 12]:

In competitive environments, agents seek a Nash equilibrium. For a two-player zero-sum Markov game with discount factor $\gamma \in (0, 1)$ and

reward sequence $\{r_t\}_{t \geq 0}$, using unconstrained parameterizations $\pi_1(\theta_1)$ and $\pi_2(\theta_2)$ of the players' policies, the value function can be written as

$$\min_{\theta_1 \in \mathbb{R}^{n_1}} \max_{\theta_2 \in \mathbb{R}^{n_2}} \Phi(\theta_1, \theta_2) = \mathbb{E}_{\pi_1(\theta_1), \pi_2(\theta_2)} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right],$$

where $\mathbb{E}_{\pi_1(\theta_1), \pi_2(\theta_2)}$ denotes expectation with respect to the probability measure induced by the initial state distribution, the transition kernel of the Markov game, and the joint policy $(\pi_1(\theta_1), \pi_2(\theta_2))$. Under the convex-concave construction used in the numerical experiments, the associated first-order saddle operator is monotone, and its optimality system can be represented as an NME in the joint parameter vector

$$x = (\theta_1, \theta_2) \in \mathbb{R}^{n_1+n_2}.$$

- **Robust Adversarial Learning (RAL)** [11, 14]:

To improve robustness against adversarial perturbations, models can be trained through problems of the form

$$\min_{\theta \in \mathbb{R}^{n_\theta}} \sum_{i=1}^N \max_{\delta_i \in \mathbb{R}^{n_\delta}} \Phi_i(\theta, \delta_i),$$

where $\theta \in \mathbb{R}^{n_\theta}$ denotes the model parameter vector, $\delta_i \in \mathbb{R}^{n_\delta}$ represents an adversarial perturbation associated with the i -th sample, and N denotes the number of samples or adversarial variables. The corresponding convex-concave constructions lead to monotone saddle operators. In particular, the rank-deficient instances used in the numerical experiments provide examples for which monotonicity holds without strong monotonicity.

- **Generative Adversarial Networks (GANs)** [1, 4]:

Let G_{θ_G} and D_{θ_D} denote the generator and discriminator parameterized by $\theta_G \in \mathbb{R}^{n_G}$ and $\theta_D \in \mathbb{R}^{n_D}$, respectively. Let p_{data} denote the probability distribution of the observed data, let $\xi \sim p_{\text{data}}$ denote a data sample, and let p_z denote a probability distribution on the latent space. The standard GAN objective can be written as

$$\min_{\theta_G \in \mathbb{R}^{n_G}} \max_{\theta_D \in \mathbb{R}^{n_D}} \Phi(\theta_G, \theta_D),$$

where

$$\Phi(\theta_G, \theta_D) = \mathbb{E}_{\xi \sim p_{\text{data}}} [\log D_{\theta_D}(\xi)] + \mathbb{E}_{z \sim p_z} [\log(1 - D_{\theta_D}(G_{\theta_G}(z)))].$$

Here, $\mathbb{E}_{\xi \sim p_{\text{data}}}[\cdot]$ denotes expectation with respect to the data distribution p_{data} , while $\mathbb{E}_{z \sim p_z}[\cdot]$ denotes expectation with respect to the latent distribution p_z . For the convex–concave categorical GAN models used in the supplementary numerical experiments, the associated first-order operator is monotone and the optimality system can be represented as an NME in the joint parameter vector

$$x = (\theta_G, \theta_D) \in \mathbb{R}^{n_G + n_D}.$$

More general nonconvex–nonconcave GAN models lie outside the monotone setting considered in this work.

This supplementary numerical material below specializes these three general application classes to explicitly generated finite-dimensional monotone equations. The constructions, parameter ranges, known solutions, Lipschitz constants, and validation procedures are given in Section 3. Thus, the presentation proceeds from the general nonlinear monotone equation (1), to its convex–concave min–max origin, and finally to the concrete learning-based test problems used in the experiments.

2 Asymptotic and Canonical OGDA-Type Direction Structures

This section examines the direction structure underlying a class of OGDA-type methods and relates it to the descent mechanisms of the proposed projection-based subspace framework. For a self-contained presentation, we also recall and redefine from the main paper the notation, subspace-direction formulas, scaling and angle conditions, and auxiliary properties that are required in the analysis below. The section is organized into three parts. First, several representative fixed step-size and related schemes for solving monotone equations are reviewed together with their known convergence properties. Second, the subspace directions generated by the proposed framework are shown to admit, along any subsequence converging to a solution, an asymptotic residual–memory representation characteristic of OGDA-type updates. Third, the update directions generated by classical OGDA-type methods are examined and shown to satisfy the same canonical asymptotic form whenever convergence occurs.

Although these methods are generally analyzed separately and may possess different convergence rates in their original formulations, their reinterpretation through the direction structure of the proposed framework places them within a common asymptotic directional framework. Consequently, the same worst-case upper-bound framework can be applied whenever the assumptions and descent conditions required by the PLS-S analysis are satisfied. For reference, Table 1 summarizes the OGDA-type schemes considered in this section, together with their step-size regimes and convergence rates reported in the literature.

For completeness, we now recall and redefine the notation and auxiliary relations from the main paper [9] that are used throughout this section. We write

$$X^* := \{x \in \mathbb{R}^n : E(x) = 0\},$$

and use (A_{lip}) to denote the Lipschitz-continuity assumption: there exists $L > 0$ such that

$$\|E(x) - E(y)\| \leq L\|x - y\|, \quad \forall x, y \in \mathbb{R}^n.$$

For the trial point and the memory vectors used in the subspace directions, we set

$$\bar{x}_\ell := x_\ell + \mu_\ell p_\ell^{\text{sc}}, \quad \mu_\ell > 0, \quad (3)$$

and define the auxiliary quantities

$$y_{\ell-1} = E(\bar{x}_{\ell-1}) - E(x_{\ell-1}), \quad s_{\ell-1} = \bar{x}_{\ell-1} - x_{\ell-1} = \mu_{\ell-1} p_{\ell-1}^{\text{sc}}. \quad (4)$$

The Jacobian-free subspace directions introduced in the main paper are

$$p_\ell = \begin{cases} \text{span}(E(x_\ell), p_{\ell-1}) & = -E(x_\ell) + \beta_\ell p_{\ell-1} & (\text{JFS2dir}), \\ \text{span}(E(x_\ell), p_{\ell-1}, y_{\ell-1}) & = -E(x_\ell) + \beta_\ell^1 p_{\ell-1} + \beta_\ell^2 y_{\ell-1} & (\text{JFS3dir1}), \\ \text{span}(E(x_\ell), y_{\ell-1}, s_{\ell-1}) & = -E(x_\ell) + \beta_\ell^3 y_{\ell-1} + \beta_\ell^4 s_{\ell-1} & (\text{JFS3dir2}), \\ \text{span}(E(x_\ell), p_{\ell-1}, h_\ell) & = -E(x_\ell) + \beta_\ell p_{\ell-1} + \beta_\ell^5 h_\ell & (\text{JFS3dir3}). \end{cases} \quad (5)$$

The updating coefficients in (5) are safeguarded and satisfy the uniform boundedness condition

$$\beta_\ell, \beta_\ell^i \in [\beta_{\min}, \beta_{\max}], \quad \text{for all } \ell \text{ and } i = 1, \dots, 5, \quad (6)$$

where

$$-\infty < \beta_{\min} \leq \beta_{\max} < \infty.$$

Before scaling, the candidate direction is required to satisfy the residual-based angle condition

$$\frac{(p_\ell)^T E(x_\ell)}{\|p_\ell\| \|E(x_\ell)\|} \leq -\Delta_a < 0, \quad \text{with } 0 < \Delta_a < 1. \quad (7)$$

If the candidate direction does not satisfy (7), assuming $E(x_\ell) \neq 0$ and $p_\ell \neq 0$, we define

$$\begin{aligned} \sigma_1 &:= E(x_\ell)^T E(x_\ell) > 0, \quad \sigma_2 := (p_\ell)^T p_\ell > 0, \\ \sigma &:= E(x_\ell)^T p_\ell, \quad \bar{\sigma} := \frac{\sigma}{\sqrt{\sigma_1 \sigma_2}} \in [-1, 1] \end{aligned} \quad (8)$$

and compute

$$t_\ell := \frac{\sigma + \Delta_a \sqrt{w}}{\sigma_1}, \quad w := \frac{\sigma_1 \sigma_2 (1 - \bar{\sigma}^2)}{1 - \Delta_a^2}. \quad (9)$$

When $w > 0$, the angle-enforced direction is

$$p_\ell^{\text{md}} := p_\ell - t_\ell E(x_\ell). \quad (10)$$

If $w = 0$, we set $p_\ell^{\text{md}} := -E(x_\ell)$. When the modification is applied, we set $p_\ell := p_\ell^{\text{md}}$; if the candidate direction already satisfies (7), p_ℓ is left unchanged. After the angle condition has been enforced, the resulting nonzero direction p_ℓ is scaled according to

$$p_\ell^{\text{sc}} := \eta_\ell p_\ell, \quad \eta_\ell := \frac{\text{mid} \{ \zeta \|E(x_\ell)\|, \|p_\ell\|, \|E(x_\ell)\| \}}{\|p_\ell\|}. \quad (11)$$

Consequently,

$$\zeta \|E(x_\ell)\| \leq \|p_\ell^{\text{sc}}\| \leq \|E(x_\ell)\|. \quad (12)$$

Since $\eta_\ell > 0$, this scaling does not change the angle between the direction and $E(x_\ell)$; hence, the scaled direction p_ℓ^{sc} also satisfies (7).

For the structural OGDA-type result below, we additionally assume, only when explicitly stated, that the scaling factors satisfy

$$0 < \eta_\ell \leq \eta_{\max} < \infty \quad (13)$$

for all sufficiently large ℓ . This additional assumption is not part of the global convergence analysis in the main paper [9]. Finally, for an infinite sequence

generated by the proposed projection framework, the asymptotic regularity established in the main paper gives

$$\lim_{\ell \rightarrow \infty} \|x_\ell - \bar{x}_\ell\| = 0, \quad \lim_{\ell \rightarrow \infty} \|x_{\ell+1} - x_\ell\| = 0. \quad (14)$$

The definitions and properties (3)–(14) are recalled here solely to support the supplementary structural analysis; their derivation and their role in the global convergence and convergence-rate analysis are given in the main paper.

Table 1: Comparison of **OGDA**-type methods and convergence rates. Here, fixed step sizes are typically chosen in terms of the Lipschitz constant L of E , while some methods may also employ iteration-dependent momentum parameters.

Method	Step-size	Rate type	Rate
EG	Fixed	Best iterate	$O(\ell^{-1/2})$
OGDA	Fixed	Best iterate	$O(\ell^{-1/2})$
EAG	Variable/anchored parameters	Last iterate	$O(\ell^{-1})$
Fast OGDA	Parameterized	Last iterate	$o(\ell^{-1})$

2.1 Representative OGDA-type methods

We briefly recall several well-known fixed step-size and related schemes for solving monotone equations together with their established convergence guarantees. Throughout this subsection, E is assumed to be monotone and L -Lipschitz continuous, and $X^* \neq \emptyset$.

- **Extragradient (EG)** [10]:

$$w_\ell = x_\ell - sE(x_\ell), \quad x_{\ell+1} = x_\ell - sE(w_\ell), \quad (15)$$

where $0 < s < 1/L$. Under these assumptions, the extragradient method is globally convergent for any initial point $x_0 \in \mathbb{R}^n$. Moreover, it is known that the residual sequence satisfies the best iterate sublinear rate, see, e.g., [5]: $\min_{1 \leq k \leq \ell} \|E(x_k)\| = O(\ell^{-1/2})$ as $\ell \rightarrow \infty$.

- **Optimistic Gradient Descent–Ascent (OGDA)** [13]:

$$x_{\ell+1} = x_\ell - 2sE(x_\ell) + sE(x_{\ell-1}), \quad (16)$$

with $0 < s < 1/(2L)$. Under the assumptions imposed in the cited convergence result, global convergence of the method is guaranteed. Furthermore, under the step-size and other assumptions imposed in [3], the best-iterate residual satisfies $\min_{1 \leq k \leq \ell} \|E(x_k)\| = O(\ell^{-1/2})$ as $\ell \rightarrow \infty$. Pointwise convergence rates are generally not available without additional assumptions.

- **Extra Anchored Gradient (EAG)** [18]:

$$w_\ell = x_\ell + \theta_\ell(x_0 - x_\ell) - sE(x_\ell), \quad x_{\ell+1} = x_\ell + \theta_\ell(x_0 - x_\ell) - sE(w_\ell), \quad (17)$$

where $\theta_\ell = O(1/\ell)$ and the step sizes are chosen according to the rules specified in the original reference. Under the assumptions imposed there, global convergence is guaranteed, and the residual norm converges at the faster last iterate rate $\|E(x_\ell)\| = O(\ell^{-1})$ as $\ell \rightarrow \infty$.

- **Fast OGDA** [2]:

$$\begin{aligned} w_\ell &= x_\ell + \theta_\ell(x_\ell - x_{\ell-1}) - s\eta_\ell E(w_{\ell-1}), \\ x_{\ell+1} &= w_\ell - s\kappa_\ell(E(w_\ell) - E(w_{\ell-1})) \end{aligned} \quad (18)$$

with suitably chosen parameter sequences θ_ℓ , η_ℓ , and κ_ℓ . Under the assumptions imposed in the cited result, the method converges globally and satisfies the rate $\|E(x_\ell)\| = o(\ell^{-1})$ as $\ell \rightarrow \infty$.

These methods differ significantly at the algorithmic level and in their specific convergence guarantees. Nevertheless, as shown in the next subsection, the subspace directions of PLS-S/PF-S admit a common asymptotic residual-memory structure analogous to OGDA-type updates.

2.2 Asymptotic OGDA-type structure of subspace directions

The subspace directions introduced in (5) are defined algebraically as derivative-free conjugate-gradient-type updates. While these directions are motivated by finite-dimensional subspace constructions, their role in the convergence analysis becomes transparent only after global convergence and asymptotic regularity of the iterates have been established. In particular, once the iterates approach the solution set, the subspace directions exhibit a specific

residual-memory structure analogous to optimistic gradient descent-ascent (OGDA) schemes. Before stating the corresponding structural result, we recall the asymptotic regularity of the iterates and establish a bound on the magnitude of the angle-enforcement correction.

Lemma 1 (Well-definedness and control of coefficient t_ℓ). *Let $\sigma_1, \sigma_2, \sigma, \bar{\sigma}$ be defined as in (8), and let t_ℓ and w be defined by (9). Suppose that p_ℓ does not satisfy the angle condition (7). Then t_ℓ is well-defined, finite, and strictly positive, and*

$$0 < t_\ell \leq \frac{\|p_\ell\|}{\|E(x_\ell)\|} \left(1 + \frac{\Delta_a}{\sqrt{1 - \Delta_a^2}} \right), \quad 0 < \Delta_a < 1. \quad (19)$$

Equivalently,

$$\|t_\ell E(x_\ell)\| \leq \left(1 + \frac{\Delta_a}{\sqrt{1 - \Delta_a^2}} \right) \|p_\ell\|. \quad (20)$$

Proof. Since $E(x_\ell) \neq 0$ and $p_\ell \neq 0$, we have

$$\sigma_1 = \|E(x_\ell)\|^2 > 0, \quad \sigma_2 = \|p_\ell\|^2 > 0.$$

From (8), $\bar{\sigma} \in [-1, 1]$, and hence $1 - \bar{\sigma}^2 \geq 0$. Since $0 < \Delta_a < 1$, we also have $1 - \Delta_a^2 > 0$. Therefore,

$$w = \frac{\sigma_1 \sigma_2 (1 - \bar{\sigma}^2)}{1 - \Delta_a^2} \geq 0,$$

so $\sqrt{w}$ is well defined. Because p_ℓ does not satisfy (7), $\bar{\sigma} > -\Delta_a$. Using $\sigma = \bar{\sigma} \sqrt{\sigma_1 \sigma_2}$, we obtain

$$\sigma + \Delta_a \sqrt{w} = \sqrt{\sigma_1 \sigma_2} \left(\bar{\sigma} + \Delta_a \sqrt{\frac{1 - \bar{\sigma}^2}{1 - \Delta_a^2}} \right).$$

If $\bar{\sigma} \geq 0$, the expression in parentheses is positive. If $-\Delta_a < \bar{\sigma} < 0$, then $\bar{\sigma}^2 < \Delta_a^2$, and consequently

$$\Delta_a \sqrt{\frac{1 - \bar{\sigma}^2}{1 - \Delta_a^2}} > -\bar{\sigma}.$$

Hence, $\sigma + \Delta_a \sqrt{w} > 0$. Since $\sigma_1 > 0$, it follows that $t_\ell > 0$.

Next, by the Cauchy–Schwarz inequality, $|\sigma| \leq \sqrt{\sigma_1 \sigma_2}$. Furthermore,

$$\sqrt{w} = \sqrt{\frac{\sigma_1 \sigma_2 (1 - \bar{\sigma}^2)}{1 - \Delta_a^2}} \leq \frac{\sqrt{\sigma_1 \sigma_2}}{\sqrt{1 - \Delta_a^2}}.$$

Therefore,

$$t_\ell \leq \frac{|\sigma|}{\sigma_1} + \frac{\Delta_a \sqrt{w}}{\sigma_1} \leq \frac{\sqrt{\sigma_1 \sigma_2}}{\sigma_1} \left(1 + \frac{\Delta_a}{\sqrt{1 - \Delta_a^2}} \right).$$

Since $\sqrt{\sigma_1 \sigma_2} = \|E(x_\ell)\| \|p_\ell\|$, we obtain

$$t_\ell \leq \frac{\|p_\ell\|}{\|E(x_\ell)\|} \left(1 + \frac{\Delta_a}{\sqrt{1 - \Delta_a^2}} \right),$$

which proves (19). Multiplying by $\|E(x_\ell)\| > 0$ gives

$$\|t_\ell E(x_\ell)\| = t_\ell \|E(x_\ell)\| \leq \left(1 + \frac{\Delta_a}{\sqrt{1 - \Delta_a^2}} \right) \|p_\ell\|,$$

which proves (20). □ □

Lemma 2 (Asymptotic regularity). *Suppose that the assumptions required for the global convergence analysis of the proposed projection framework hold. Then either the method terminates in a finite number of iterations or it generates an infinite sequence $\{x_\ell\}_{\ell \geq 0}$ satisfying*

$$\lim_{\ell \rightarrow \infty} \|x_\ell - \bar{x}_\ell\| = 0, \quad \lim_{\ell \rightarrow \infty} \|x_{\ell+1} - x_\ell\| = 0.$$

In particular, the iterates are asymptotically regular.

These properties are used explicitly in the asymptotic OGDA-type analysis below. The following result makes the connection with OGDA-type schemes precise by showing that, along any subsequence converging to a solution, each scaled subspace direction admits an OGDA-type asymptotic expansion with an asymptotically vanishing remainder.

Theorem 1 (Uniform asymptotic OGDA-type structure). *Let Assumption (A_{lip}) hold and suppose that the angle condition (7) is satisfied at every iteration. Assume further that the subspace coefficients satisfy the boundedness condition (6), that the scaling factors satisfy (11) and there exist constants*

$0 < \eta_{\max} < \infty$ such that (13) holds, for all $\ell \geq 0$, and that the auxiliary sequence $\{h_\ell\}_{\ell \geq 0}$ satisfies $\|h_\ell\| = O(\|E(x_{\ell-1})\|)$. Let $\{x_\ell\}_{\ell \geq 0}$ be generated by PLS-S, and let $\{p_\ell^{\text{sc}}\}_{\ell \geq 0}$ be the scaled directions defined by (11). Then, for any subsequence $\{\ell_j\}_{j \geq 0}$ such that $x_{\ell_j} \rightarrow x^*$ with $x^* \in X^*$, the corresponding directions admit the representation

$$p_{\ell_j}^{\text{sc}} = -E(x_{\ell_j}) + \alpha_{\ell_j} E(x_{\ell_j-1}) + r_{\ell_j},$$

where $\{\alpha_\ell\}_{\ell \geq 0}$ is uniformly bounded and $\|r_{\ell_j}\| = O(\|E(x_{\ell_j-1})\|) + o(1)$.

Proof. We proceed in justified steps.

Step 1: Unified algebraic representation. All subspace directions admit the representation

$$p_\ell = -E(x_\ell) + \beta_\ell^{(p)} p_{\ell-1}^{\text{sc}} + \beta_\ell^{(y)} y_{\ell-1} + \beta_\ell^{(s)} s_{\ell-1} + \beta_\ell^{(h)} h_\ell. \quad (21)$$

Here, coefficients corresponding to terms that do not occur in a particular direction are set equal to zero. Substituting $y_{\ell-1}$ and $s_{\ell-1}$ (defined in (4)) yields

$$p_\ell = -E(x_\ell) + \beta_\ell^{(p)} p_{\ell-1}^{\text{sc}} + \beta_\ell^{(y)} (E(\bar{x}_{\ell-1}) - E(x_{\ell-1})) + \beta_\ell^{(s)} \mu_{\ell-1} p_{\ell-1}^{\text{sc}} + \beta_\ell^{(h)} h_\ell.$$

Equivalently,

$$p_\ell = -E(x_\ell) - \beta_\ell^{(y)} E(x_{\ell-1}) + (\beta_\ell^{(p)} + \beta_\ell^{(s)} \mu_{\ell-1}) p_{\ell-1}^{\text{sc}} + \beta_\ell^{(y)} E(\bar{x}_{\ell-1}) + \beta_\ell^{(h)} h_\ell.$$

If the angle-enforcement modification (10) is performed, that is, $p_\ell^{\text{md}} = p_\ell - t_\ell E(x_\ell)$, with t_ℓ determined by (9), then the resulting direction retains the same residual-memory structure. Indeed, the additional term is a multiple of the current residual, and Lemma 1 gives (20). To simplify the notation, we continue to denote the modified direction p_ℓ^{md} by p_ℓ .

Step 2: Incorporating scaling. Using (11), $p_\ell^{\text{sc}} = \eta_\ell p_\ell$, and multiplying the representation obtained in Step 1 by η_ℓ , we obtain $p_\ell^{\text{sc}} = -E(x_\ell) + \alpha_\ell E(x_{\ell-1}) + r_\ell$ and $\alpha_\ell := -\eta_\ell \beta_\ell^{(y)}$, where

$$\begin{aligned} r_\ell &= \eta_\ell (\beta_\ell^{(p)} + \beta_\ell^{(s)} \mu_{\ell-1}) p_{\ell-1}^{\text{sc}} + \eta_\ell \beta_\ell^{(y)} E(\bar{x}_{\ell-1}) \\ &\quad + \eta_\ell \beta_\ell^{(h)} h_\ell + (1 - \eta_\ell (1 + t_\ell)) E(x_\ell). \end{aligned} \quad (22)$$

If no angle-enforcement modification is required, we set $t_\ell = 0$, in which case the last term in r_ℓ reduces to $(1 - \eta_\ell) E(x_\ell)$.

Step 3: Uniform boundedness. By (6), there exists $\bar{\beta} > 0$ such that $|\beta_\ell^{(i)}| \leq \bar{\beta}$ for all ℓ and all relevant indices i . Since η_ℓ satisfies (13) and $\alpha_\ell = -\eta_\ell \beta_\ell^{(y)}$, $|\alpha_\ell| \leq \eta_{\max} \bar{\beta}$, and hence the sequence $\{\alpha_\ell\}_{\ell \geq 0}$ is uniformly bounded.

Step 4: Control of the memory magnitude. From the upper scaling bound (12), $\|p_{\ell-1}^{\text{sc}}\| \leq \|E(x_{\ell-1})\|$. Consequently, $\|p_{\ell-1}^{\text{sc}}\| = O(\|E(x_{\ell-1})\|)$.

Step 5: Lipschitz control. Let $M_\mu := \sup_{\ell \geq 0} \mu_\ell < \infty$. Indeed, one may take $M_\mu = \mu_{\text{init}}$ in the line-search variant and $M_\mu = \alpha$ in the fixed step-size variant. Using Step 4 and (3), we obtain

$$\|\bar{x}_{\ell-1} - x_{\ell-1}\| = \mu_{\ell-1} \|p_{\ell-1}^{\text{sc}}\| \leq M_\mu \|E(x_{\ell-1})\|.$$

Using the Lipschitz continuity of E , it follows that

$$\|E(\bar{x}_{\ell-1})\| \leq \|E(x_{\ell-1})\| + L \|\bar{x}_{\ell-1} - x_{\ell-1}\| \leq (1 + LM_\mu) \|E(x_{\ell-1})\|.$$

Moreover, using the Lipschitz continuity of E and Lemma 2(ii-2),

$$\|E(x_\ell) - E(x_{\ell-1})\| \leq L \|x_\ell - x_{\ell-1}\| \rightarrow 0.$$

Hence, $E(x_\ell) = E(x_{\ell-1}) + o(1)$.

Step 6: Remainder estimate. Recall r_ℓ from (22). Then, we estimate each term separately.

- (i) By Step 4, $\|p_{\ell-1}^{\text{sc}}\| \leq \|E(x_{\ell-1})\|$. Since $0 < \eta_\ell \leq \eta_{\max}$ and $\beta_\ell^{(p)}$, $\beta_\ell^{(s)}$, and $\mu_{\ell-1}$ are uniformly bounded, we obtain

$$\left\| \eta_\ell (\beta_\ell^{(p)} + \beta_\ell^{(s)} \mu_{\ell-1}) p_{\ell-1}^{\text{sc}} \right\| \leq C_i \|E(x_{\ell-1})\|.$$

- (ii) By Step 5, $\|E(\bar{x}_{\ell-1})\| \leq (1 + LM_\mu) \|E(x_{\ell-1})\|$. Using $0 < \eta_\ell \leq \eta_{\max}$ and the boundedness of $\beta_\ell^{(y)}$, $\|\eta_\ell \beta_\ell^{(y)} E(\bar{x}_{\ell-1})\| \leq C_{ii} \|E(x_{\ell-1})\|$.

- (iii) By assumption, $\|h_\ell\| = O(\|E(x_{\ell-1})\|)$, and hence $\|\eta_\ell \beta_\ell^{(h)} h_\ell\| = O(\|E(x_{\ell-1})\|)$.

- (iv) By Step 5, $\|E(x_\ell)\| \leq \|E(x_{\ell-1})\| + o(1)$. Since $0 < \eta_\ell \leq \eta_{\max}$,

$$\|(1 - \eta_\ell) E(x_\ell)\| \leq C_{iv} \|E(x_{\ell-1})\| + o(1).$$

Whenever the angle-enforcement modification is applied, (11), (12), and (20) yields

$$\begin{aligned}\|t_\ell \eta_\ell E(x_\ell)\| &\leq \left(1 + \frac{\Delta_a}{\sqrt{1 - \Delta_a^2}}\right) \|p_\ell^{\text{sc}}\| \leq \left(1 + \frac{\Delta_a}{\sqrt{1 - \Delta_a^2}}\right) \|E(x_\ell)\| \\ &= O(\|E(x_{\ell-1})\|) + o(1).\end{aligned}$$

Collecting the above estimates and using the triangle inequality, there exists a constant $C > 0$ such that $\|r_\ell\| \leq C\|E(x_{\ell-1})\| + o(1)$. Restricting the estimate to the convergent subsequence $\{\ell_j\}_{j \geq 0}$, $E(x_{\ell_j}) \rightarrow E(x^*) = 0$. Moreover, by asymptotic regularity and the Lipschitz continuity of E ,

$$\|E(x_{\ell_j}) - E(x_{\ell_j-1})\| \leq L\|x_{\ell_j} - x_{\ell_j-1}\| \rightarrow 0.$$

It follows that $E(x_{\ell_j-1}) \rightarrow 0$. Combining this fact with $\|r_\ell\| \leq C\|E(x_{\ell-1})\| + o(1)$, $\|r_{\ell_j}\| \leq C\|E(x_{\ell_j-1})\| + o(1)$. So, $\|r_{\ell_j}\| = O(\|E(x_{\ell_j-1})\|) + o(1)$, which completes the proof. $\square$

This additional assumption on the scaling factors is used only for the structural OGDA-type interpretation and is not required by the global convergence analysis of PLS-S/PF-S.

2.3 Unification of OGDA-Type Methods within the proposed PLS-S/PF-S Framework

The asymptotic OGDA-type structure established in 1 is not unique to the subspace directions generated by PLS-S. In this subsection, we show that several classical fixed step-size methods for monotone equations generate update directions that admit the same residual-memory structure. This allows us to interpret these methods within a common asymptotic directional framework and to identify the additional descent conditions required to obtain convergence-rate bounds analogous to those established for PLS-S.

2.3.1 Canonical asymptotic OGDA-type direction structure

Theorem 2 (Asymptotic OGDA-type direction structure). *Let Assumption (A_{lip}) hold, and let $\{x_\ell\}_{\ell \geq 0}$ be a sequence generated by one of the methods (15)–(17).*

Assume that $\{x_{\ell_j}\}_{j \geq 0}$ is a subsequence converging to a solution $x^* \in X^*$ and that, for all sufficiently large indices, $\|x_{\ell+1} - x_\ell\| = O(\|E(x_\ell)\|)$. Then, along this subsequence, the associated update directions can be written as

$$p_{\ell_j} = -E(x_{\ell_j}) + \alpha_{\ell_j} E(x_{\ell_j-1}) + r_{\ell_j}, \quad \|r_{\ell_j}\| = O(\|E(x_{\ell_j-1})\|) + o(1), \quad (23)$$

where $\{\alpha_\ell\}_{\ell \geq 0}$ is a bounded scalar sequence.

Proof. Let $x_{\ell_j} \rightarrow x^* \in X^*$. By continuity of E , $E(x_{\ell_j}) \rightarrow E(x^*) = 0$. Moreover, using the Lipschitz continuity of E and the increment condition, there exists a constant $C_{\text{inc}} > 0$ such that, for all sufficiently large indices,

$$\|E(x_\ell) - E(x_{\ell-1})\| \leq L\|x_\ell - x_{\ell-1}\| \leq LC_{\text{inc}}\|E(x_{\ell-1})\|.$$

Hence,

$$\|E(x_\ell)\| \leq (1 + LC_{\text{inc}})\|E(x_{\ell-1})\| \quad (24)$$

for all sufficiently large ℓ . In particular,

$$\|E(x_{\ell_j})\| = O(\|E(x_{\ell_j-1})\|). \quad (25)$$

In each case below, α_ℓ denotes the coefficient of $E(x_{\ell-1})$ in the corresponding decomposition.

1. Extragradient (EG). From (15), $p_\ell = -E(w_\ell)$ and $w_\ell = x_\ell - sE(x_\ell)$. Therefore,

$$p_\ell = -E(x_\ell) + r_\ell, \quad r_\ell := E(x_\ell) - E(w_\ell).$$

By the Lipschitz continuity of E , $\|r_\ell\| = \|E(x_\ell) - E(w_\ell)\| \leq Ls\|E(x_\ell)\|$. Here, $\alpha_\ell = 0$. Thus, along the subsequence, (25) gives $\|r_{\ell_j}\| \leq Ls\|E(x_{\ell_j})\| = O(\|E(x_{\ell_j-1})\|)$.

2. OGDA. From (16), after dividing the update by the fixed step size s , the associated direction is $p_\ell = -2E(x_\ell) + E(x_{\ell-1})$. Hence, $p_\ell = -E(x_\ell) + E(x_{\ell-1}) + r_\ell$, $r_\ell := -E(x_\ell)$. Thus, $\alpha_\ell = 1$. Using (25), we obtain $\|r_{\ell_j}\| = \|E(x_{\ell_j})\| = O(\|E(x_{\ell_j-1})\|)$.

3. Extra Anchored Gradient (EAG). From (17), after dividing the update by the fixed step size s , the associated direction is $p_\ell = -E(w_\ell) + (\theta_\ell/s)(x_0 - x_\ell)$. Therefore,

$$p_\ell = -E(x_\ell) + r_\ell,$$

where $r_\ell := E(x_\ell) - E(w_\ell) + (\theta_\ell/s)(x_0 - x_\ell)$. Here, $\alpha_\ell = 0$. Since the subsequence $\{x_{\ell_j}\}_{\ell_j \geq 0}$ converges, it is bounded, and hence $\|x_0 - x_{\ell_j}\| = O(1)$. Moreover,

$$w_\ell - x_\ell = \theta_\ell(x_0 - x_\ell) - sE(x_\ell).$$

By the Lipschitz continuity of E ,

$$\|E(w_\ell) - E(x_\ell)\| \leq L\|w_\ell - x_\ell\| \leq L\theta_\ell\|x_0 - x_\ell\| + Ls\|E(x_\ell)\|.$$

Consequently, along the convergent subsequence, $\|r_{\ell_j}\| = O(\theta_{\ell_j}) + O(\|E(x_{\ell_j})\|)$. Since $\theta_{\ell_j} \rightarrow 0$, and by (25), $\|r_{\ell_j}\| = O(\|E(x_{\ell_j-1})\|) + o(1)$. Combining the three cases yields (23). $\square$

4. Fast OGDA. The fast OGDA iteration contains inertial, delayed-residual, and residual-difference terms. Its precise reduction to the canonical form (23) depends on the parameter identities imposed in the original method. We therefore do not claim the representation (23) for fast OGDA without those additional identities. Its known last-iterate convergence result remains method-specific.

2.3.2 Unified convergence-rate implications

The canonical representation (23) shows that, along any subsequence converging to a solution, the update directions of the methods (15)–(17) admit an asymptotic residual–memory decomposition analogous to that of the PLS–S directions.

This representation is, however, an algebraic and asymptotic property. It does not by itself imply a uniform angle condition, a sufficient-decrease inequality, or the Fejér monotonicity used in the analysis of PLS–S. Consequently, the convergence-rate results of Section 6 in [9] do not automatically transfer to the methods (15)–(18). The following result is independent of Theorem 2 and specifies an additional descent condition under which the rates of Section 6 in [9] are valid.

Corollary 1 (Local rates under a Fejér-type residual decrease). *Suppose that one of the methods (15)–(18) generates a sequence $\{x_\ell\}_{\ell \geq 0}$ converging to $x^* \in X^*$. Assume that there exist constants $c > 0$ and $\ell_0 \geq 0$ such that*

$$\text{dist}(x_{\ell+1}, X^*)^2 \leq \text{dist}(x_\ell, X^*)^2 - c\|E(x_\ell)\|^2 \quad (26)$$

for every $\ell \geq \ell_0$. Then $\min_{\ell_0 \leq k \leq \ell} \|E(x_k)\| = O(\ell^{-1/2})$. If, in addition, a local error bound holds at x^* , namely, there exist $\vartheta > 0$ and a neighborhood $\mathcal{N}$ of x^* such that

$$\vartheta \operatorname{dist}(x, X^*) \leq \|E(x)\| \quad \text{for all } x \in \mathcal{N},$$

then there exist constants $C > 0$ and $q \in (0, 1)$ such that $\operatorname{dist}(x_\ell, X^*) \leq Cq^\ell$ for all sufficiently large ℓ .

Proof. Summing (26) from $\ell = \ell_0$ to $\ell = N$ yields

$$c \sum_{\ell=\ell_0}^N \|E(x_\ell)\|^2 \leq \operatorname{dist}(x_{\ell_0}, X^*)^2 - \operatorname{dist}(x_{N+1}, X^*)^2 \leq \operatorname{dist}(x_{\ell_0}, X^*)^2.$$

Therefore, $\sum_{\ell=\ell_0}^\infty \|E(x_\ell)\|^2 < \infty$. For every $N \geq \ell_0$,

$$(N - \ell_0 + 1) \min_{\ell_0 \leq k \leq N} \|E(x_k)\|^2 \leq \sum_{\ell=\ell_0}^N \|E(x_\ell)\|^2 \leq \frac{1}{c} \operatorname{dist}(x_{\ell_0}, X^*)^2.$$

Hence, $\min_{\ell_0 \leq k \leq N} \|E(x_k)\| \leq \frac{\operatorname{dist}(x_{\ell_0}, X^*)}{\sqrt{c(N - \ell_0 + 1)}}$, which gives $\min_{\ell_0 \leq k \leq \ell} \|E(x_k)\| =$

$O(\ell^{-1/2})$. Now suppose that the local error bound holds. Since $x_\ell \rightarrow x^*$, there exists $\ell_1 \geq \ell_0$ such that $x_\ell \in \mathcal{N}$ for all $\ell \geq \ell_1$. Thus, $\|E(x_\ell)\| \geq \vartheta \operatorname{dist}(x_\ell, X^*)$ for all $\ell \geq \ell_1$. Combining this inequality with (26), we obtain

$$\operatorname{dist}(x_{\ell+1}, X^*)^2 \leq \operatorname{dist}(x_\ell, X^*)^2 - c\|E(x_\ell)\|^2 \leq (1 - c\vartheta^2) \operatorname{dist}(x_\ell, X^*)^2$$

for every $\ell \geq \ell_1$. If $c\vartheta^2 \geq 1$, then finite convergence occurs after entering the local neighborhood. Otherwise, $0 < c\vartheta^2 < 1$. Setting $q := \sqrt{1 - c\vartheta^2} \in (0, 1)$, we obtain

$$\operatorname{dist}(x_{\ell+1}, X^*) \leq q \operatorname{dist}(x_\ell, X^*) \quad \text{for all } \ell \geq \ell_1.$$

Therefore, $\operatorname{dist}(x_\ell, X^*) \leq \operatorname{dist}(x_{\ell_1}, X^*)q^{\ell-\ell_1}$ for all $\ell \geq \ell_1$, which proves the stated R -linear convergence of the distance to the solution set. $\square$

3 Learning-Based Test Problems

To complement the standard test set, we consider 100 learning-based test problems generated from the three application classes discussed in the introduction: multi-agent reinforcement learning (MARL), robust adversarial

learning, and generative adversarial networks (GANs). Since the corresponding min–max formulations and their relation to NMEs have already been presented, we focus here on the numerical construction of the test instances.

All problems are generated using the fixed random seed 42 to ensure reproducibility. For each application class, four variants are considered. These variants modify the problem dimension and application-specific parameters and are intended to provide problems with different structural and numerical properties. Consistently with (1), throughout this subsection n denotes the dimension of the NME variable $x \in \mathbb{R}^n$. For each generated instance, we store the monotone operator E , the corresponding min–max objective Φ , the merit function

$$f(x) = \frac{1}{2} \|E(x)\|^2,$$

a Lipschitz constant L , a known solution x^* , an initial point x_0 , and finite initialization and validation ranges used only to generate representative sample points.

In Table 2, m denotes the number of actions available to each player in the MARL construction and the number of categorical outcomes in the GAN construction. The parameter $\gamma \in (0, 1)$ denotes the discount factor in the MARL problems. For the robust adversarial-learning problems, p denotes the dimension of the model parameter, N denotes the number of scalar adversarial variables, and $\delta_{\text{init}} > 0$ denotes their initialization scale. For the GAN problems, $d = (d_1, \dots, d_m)^\top \in \mathbb{R}^m$ denotes the discriminator parameter vector, and $d_{\text{init}} > 0$ denotes its initialization scale.

Table 2 summarizes the main parameters used to distinguish the four variants of each problem class.

MARL instances. For the MARL test problems, we generate discounted two-player zero-sum one-state Markov-game instances. In the numerical construction, m denotes the number of actions available to each player. The unique state transitions to itself with probability one. Let $R \in \mathbb{R}^{m \times m}$ denote the scaled payoff matrix and let $R_{\text{stage}} = (1 - \gamma)R$ denote the one-step reward matrix. Consequently, for stationary mixed policies p and q , the discounted objective reduces to $p^\top Rq$.

We use the affine reduced representation

$$S = \begin{bmatrix} I_{m-1} \\ -\mathbf{1}^\top \end{bmatrix}, \quad e_m = \begin{bmatrix} 0 & \dots & 0 & 1 \end{bmatrix}^\top,$$

Table 2: Variants used to generate the learning-based test problems.

Class	Variant	Main dimension	Key parameter	Structure
MARL	1	$m \in [5, 10]$	$\gamma = 0.90$	mild conditioning
	2	$m \in [8, 15]$	$\gamma = 0.95$	moderate conditioning
	3	$m \in [12, 20]$	$\gamma = 0.98$	ill-conditioned
	4	$m \in [18, 30]$	$\gamma = 0.99$	highly ill-conditioned
RAL	1	$p \in [10, 20], N \in [30, 50]$	$\delta_{\text{init}} = 0.25$	full rank
	2	$p \in [25, 40], N \in [15, 24]$	$\delta_{\text{init}} = 0.50$	rank deficient
	3	$p \in [40, 60], N \in [20, 35]$	$\delta_{\text{init}} = 0.75$	rank deficient
	4	$p \in [60, 90], N \in [25, 45]$	$\delta_{\text{init}} = 1.00$	rank deficient
GAN	1	$m \in [5, 10]$	$d_{\text{init}} = 6$	categorical/logistic
	2	$m \in [8, 15]$	$d_{\text{init}} = 8$	categorical/logistic
	3	$m \in [12, 20]$	$d_{\text{init}} = 10$	categorical/logistic
	4	$m \in [18, 30]$	$d_{\text{init}} = 12$	categorical/logistic

and write $p = e_m + Su$ and $q = e_m + Sv$, where $u, v \in \mathbb{R}^{m-1}$. Consequently, the dimension of the corresponding NME is $n = 2(m-1)$. The resulting bilinear objective is defined for every $(u, v) \in \mathbb{R}^{2(m-1)}$, and its saddle operator is therefore an unconstrained mapping on $\mathbb{R}^{2(m-1)}$.

The payoff matrix is randomly generated on the subspace orthogonal to the all-ones vector. In particular, its construction ensures that its row and column sums vanish. This provides a known solution corresponding to the uniform mixed policies,

$$u^* = v^* = \frac{1}{m} \mathbf{1},$$

and, at the same time, preserves the saddle-point structure required for the associated operator to be monotone. The four variants modify the number of actions, the discount factor γ , and the spectral properties of the payoff matrix. The spectral conditioning levels used in the four variants are 2, 5, 20, and 100, respectively. In particular, the values

$$\gamma \in \{0.90, 0.95, 0.98, 0.99\}$$

are used together with these increasingly demanding spectral constructions.

For the reduced bilinear coupling $C = S^\top RS$, the exact Lipschitz constant of the corresponding saddle operator is $\|C\|_2$. Hence, the value stored with each MARL instance is

$$L = \max\{\|C\|_2, \text{eps}\},$$

where $\text{eps} > 0$ is a small numerical tolerance used to prevent L from becoming zero. For initialization and numerical validation, the reduced coordinates are

sampled from

$$0 \leq x_i \leq \frac{1}{m-1}.$$

These finite ranges are used only to generate representative initial and validation points; the iterates themselves remain unconstrained.

RAL instances. For the robust adversarial-learning problems, p denotes the dimension of the model parameter $\theta \in \mathbb{R}^p$, while N denotes the number of scalar adversarial variables. The joint variable is represented as

$$x = (\theta, \delta), \quad \delta = (\delta_1, \dots, \delta_N)^\top \in \mathbb{R}^N,$$

and the dimension of the corresponding NME is therefore $n = p + N$. For each instance, a reference parameter $\theta^* \in \mathbb{R}^p$ is generated randomly as $\theta^* = 0.2z$, where z is a standard Gaussian random vector. Let $A \in \mathbb{R}^{N \times p}$ and $B \in \mathbb{R}^{p \times N}$ denote the randomly generated matrices and let $\eta = (\eta_1, \dots, \eta_N)^\top$ contain the positive curvature parameters. These parameters are generated according to

$$\eta_i = \eta_{\text{base}}(0.75 + 0.5\xi_i), \quad \xi_i \in [0, 1],$$

where $\eta_{\text{base}} \in \{0.50, 0.20, 0.05, 0.01\}$ for variants 1–4, respectively.

With $e = \theta - \theta^*$, the unconstrained convex–concave objective is

$$\Phi(\theta, \delta) = \frac{1}{2N} \|Ae\|^2 + \frac{1}{N} e^\top B\delta - \frac{1}{2N} \sum_{i=1}^N \eta_i \delta_i^2.$$

The first variant employs a full-column-rank matrix A , whereas in variants 2–4 we have $N < p$ by construction and hence $A^\top A$ is rank deficient. This distinction allows the test set to contain monotone operators that need not be strongly monotone. The initialization scale is varied according to

$$\delta_{\text{init}} \in \{0.25, 0.50, 0.75, 1.00\}.$$

The Jacobian of the resulting affine monotone operator is

$$M = \begin{bmatrix} A^\top A/N & B/N \\ -B^\top/N & \text{diag}(\eta/N) \end{bmatrix},$$

and the Lipschitz constant stored with the instance is

$$L = \max\{\|M\|_2, \text{eps}\}.$$

The known solution is

$$x^* = \begin{bmatrix} \theta^* \\ 0 \end{bmatrix}.$$

For initialization and numerical validation, the model-parameter sampling radius is selected as

$$\theta_{\text{bd}} = \max \left\{ 2, 2 \max_i |\theta_i^*| + 1 \right\},$$

and the sample points are drawn from $-\theta_{\text{bd}} \leq \theta_i \leq \theta_{\text{bd}}$ and $-\delta_{\text{init}} \leq \delta_i \leq \delta_{\text{init}}$. These finite ranges are used only for sampling and are not imposed on the iterates.

GAN instances. For the GAN problems, we use the globally defined convex–concave categorical/logistic surrogate

$$\Phi(u, d) = \frac{1}{2} \left(p_{\text{data}} - p_G(u) \right)^\top d - \sum_{j=1}^m \log \left(2 \cosh(d_j/2) \right),$$

where $u \in \mathbb{R}^{m-1}$, $d = (d_1, \dots, d_m)^\top \in \mathbb{R}^m$, and $p_G(u) = e_m + Su$. Hence, the dimension of the corresponding NME is $n = (m-1) + m = 2m-1$.

For every instance, the reference distribution $p_{\text{data}} \in \mathbb{R}^m$ is generated as a strictly positive probability vector satisfying

$$(p_{\text{data}})_j > 0, \quad j = 1, \dots, m, \quad \sum_{j=1}^m (p_{\text{data}})_j = 1.$$

More precisely, the construction starts from the uniform probability vector

$$p_{\text{base}} = \frac{1}{m} \mathbf{1}$$

and adds a zero-sum random perturbation whose magnitude is controlled by the variant-dependent parameter

$$\text{spread} \in \{0.02, 0.015, 0.010, 0.005\}.$$

The perturbation is scaled so that the resulting reference vector remains strictly positive and its first $m-1$ components lie inside the finite initialization range. In particular, the construction uses

$$p_{\min} = \frac{0.25}{m}, \quad p_{\max} = \frac{0.99}{m-1}.$$

Using

$$\sigma(t) = \frac{1}{1 + \exp(-t)},$$

the corresponding saddle operator is

$$E(u, d) = \begin{bmatrix} -\frac{1}{2}S^\top d \\ \frac{1}{2}(p_G(u) - p_{\text{data}}) + \sigma(d) - \frac{1}{2}\mathbf{1} \end{bmatrix},$$

where $\sigma(d)$ is understood componentwise. The known solution satisfies $p_G(u^*) = p_{\text{data}}$ and $d^* = 0$. Thus,

$$u^* = (p_{\text{data}})_{1:m-1}, \quad d^* = 0.$$

The symmetric part of the Jacobian of E is

$$\begin{bmatrix} 0 & 0 \\ 0 & \text{diag}(\sigma(d_j)(1 - \sigma(d_j))) \end{bmatrix} \succeq 0,$$

so the operator is monotone on all of $\mathbb{R}^{2m-1}$. Since $\|S\|_2 = \sqrt{m}$ and $0 \leq \sigma(t)(1 - \sigma(t)) \leq 1/4$, the stored global Lipschitz upper bound is

$$L = \frac{1}{2}\sqrt{m} + \frac{1}{4}.$$

The four variants change the number of categorical outcomes, the perturbation spread, and the discriminator initialization scale $d_{\text{init}} \in \{6, 8, 10, 12\}$. For initialization and numerical validation, the discriminator components are sampled from $|d_j| \leq d_{\text{init}}$ for $j = 1, \dots, m$, and the generator coordinates are sampled from

$$0 \leq u_j \leq \frac{1}{m-1}, \quad j = 1, \dots, m-1.$$

These finite ranges are used only for sampling representative initial and validation points. All logistic and softplus expressions are evaluated using numerically stable formulas.

Unconstrained domains. For all three application classes, the NME variable is defined on $\mathbb{R}^n$, and all methods are applied directly to

$$E(x) = 0, \quad x \in \mathbb{R}^n.$$

The finite ranges introduced above are used only for generating initial points and random validation samples.

Lipschitz constants. A global Lipschitz constant $L > 0$, or a valid computable global upper bound, is associated with every generated operator. For MARL, the exact value is obtained from the reduced bilinear coupling matrix; for robust adversarial learning, it is obtained from the spectral norm of the affine saddle-operator matrix; and for GANs the valid global upper bound $\frac{1}{2}\sqrt{m} + \frac{1}{4}$ is used. The value of L is stored with each instance and is supplied to algorithms whose stepsize rules require knowledge of a Lipschitz constant. In particular, the operators are not artificially rescaled merely to enforce $L = 1$; the stored constant reflects the scaling of the generated instance.

Initial points. The known solution x^* is used only to verify the correctness of the generated instance and is not used as the initial iterate. For each problem, a random starting point $x_0 \in \mathbb{R}^n$ is generated from the class-specific finite sampling ranges described above. To prevent the test problems from being initialized arbitrarily close to a solution, the generation procedure attempts to select a starting point satisfying

$$10^{-2} \leq \|E(x_0)\| \leq 1.$$

At most 2000 random candidates are tested. If no candidate satisfies the prescribed residual interval, the finite candidate having the largest residual among the tested points is retained.

Validation of the generated problems. Before an instance is included in the test set, several numerical checks are performed. First, the dimension of the operator output is verified to agree with the NME dimension n . The dimensions of x_0 and x^* are checked accordingly. All components of x_0 , x^* , $E(x_0)$, and $E(x^*)$ are also required to be finite. The known solution is checked through

$$\|E(x^*)\| \leq \tau_{\text{val}} = 10^{-9}.$$

Monotonicity is additionally checked numerically using 50 randomly generated pairs $(x, y) \in \mathbb{R}^n \times \mathbb{R}^n$ drawn from the class-specific finite validation ranges for every instance. In particular, with

$$m_{\text{mon}} := \langle E(x) - E(y), x - y \rangle,$$

the numerical monotonicity test requires

$$m_{\text{mon}} \geq -10^{-10} (1 + \|E(x) - E(y)\| \|x - y\|).$$

Similarly, the stored Lipschitz constant is checked by evaluating

$$\frac{\|E(x) - E(y)\|}{\|x - y\|} \leq L(1 + 10^{-8})$$

for every randomly sampled distinct pair. These tests are not substitutes for the analytical global monotonicity and Lipschitz properties of the underlying constructions; rather, they are used as implementation checks to detect errors in the generated matrices, scaling, parameters, or operator evaluations.

Accuracy and reporting measures. For all three application classes, the residual norm is

$$\text{Res}(x) = \|E(x)\|,$$

while the scalar merit function used by the stopping mechanism is

$$f(x) = \frac{1}{2}\|E(x)\|^2.$$

It is important to distinguish these two quantities. In the numerical experiments, the prescribed stopping tolerance $\varepsilon > 0$ is imposed on the merit function, so that an instance is declared solved when

$$f(x) = \frac{1}{2}\|E(x)\|^2 \leq \varepsilon = 10^{-5}.$$

For the representative learning-based problems considered later, we also report the error in the objective value of the corresponding min–max formulation. This provides a complementary measure of progress: the merit function $\frac{1}{2}\|E(x)\|^2$ measures accuracy for the NME actually solved by the algorithms, whereas the objective-value error measures progress in terms of the underlying learning problem.

The 100 test problems are obtained from the three application classes and their four variants described above. Thus, there are 12 possible class–variant combinations. For each generated problem, the application class is selected uniformly at random from the three classes and the variant is selected uniformly at random from the four variants. Since the random seed is fixed, both the resulting distribution among the three application classes and the individual generated instances are reproducible.

4 A comparison between the JFS2dir variants

Table 3 compares the fixed-step PF-S and line-search PLS-S variants associated with the JFS2dir directions; PLS-S-Z1 is also included as a reference variant. All reported methods are fully robust on this test set, solving all

100 problems without anomalies. Hence, their relative performance is distinguished primarily by computational efficiency. In terms of residual evaluations, the fixed-step variants PF-S-LS and PF-S-PR perform best, both attaining $e_{T,s}^{\text{nf}} = 77\%$. They are followed by PF-S-DL, PF-S-HZ, and PF-S-DY, each with $e_{T,s}^{\text{nf}} = 71\%$, and by PF-S-HS with $e_{T,s}^{\text{nf}} = 68\%$. In terms of computational time, PF-S-DY attains the largest mean efficiency, $e_{T,s}^{\text{msec}} = 68\%$, followed by PF-S-HZ with 66% and PF-S-DL with 65%. Among the JFS2dir variants, these results show that the fixed-step implementations are particularly competitive in terms of both residual evaluations and computational time, while the relative efficiency still depends on the particular direction formula and step-size strategy.

stopping test: $f \leq 1\text{e-}05,$			$\text{sec} \leq 300,$			$\text{nf} \leq 20000$	
100 of 100 problems solved							
$\text{dim} \in [8, 135]$			# of anomalies			$e_{T,s}$ in %	
solver		solved	#n	#t	#f	$e_{T,s}^{\text{nf}}$	$e_{T,s}^{\text{msec}}$
PF-S-LS	F-LS	100	0	0	0	77	63
PF-S-PR	F-PR	100	0	0	0	77	64
PF-S-DL	F-DL	100	0	0	0	71	65
PF-S-HZ	F-HZ	100	0	0	0	71	66
PF-S-DY	F-DY	100	0	0	0	71	68
PF-S-HS	F-HS	100	0	0	0	68	61
PF-S-FR	F-FR	100	0	0	0	62	59
PLS-S-PR	PR	100	0	0	0	55	41
PLS-S-LS	LS	100	0	0	0	55	34
PLS-S-Z1	Z1	100	0	0	0	55	42
PLS-S-FR	FR	100	0	0	0	49	20
PLS-S-DL	DL	100	0	0	0	49	38
PLS-S-DY	DY	100	0	0	0	49	39
PLS-S-HZ	HZ	100	0	0	0	48	38
PLS-S-HS	HS	100	0	0	0	44	29

Table 3: Comparison between the JFS2dir variants.

5 A comparison between the JFS3dir1 and JFS3dir2 variants

Table 4 compares the fixed-step PF-S and line-search PLS-S variants associated with the JFS3dir1 and JFS3dir2 directions in terms of robustness and mean efficiency. All reported variants solve all 100 test problems without anomalies and are therefore equally robust on this test set. In terms of

residual evaluations, PLS-S-Z2 performs best, attaining the highest mean efficiency $e_{T,s}^{\text{nf}} = 89\%$, followed by PLS-S-Z1 with 44% and PF-S-Z1 with 41%. Regarding computational time, PLS-S-Z2 also attains the highest mean efficiency, with $e_{T,s}^{\text{msec}} = 87\%$. Among the JFS3dir2 variants, PF-S-An1 and PF-S-An2 are the most efficient in terms of residual evaluations, both with $e_{T,s}^{\text{nf}} = 39\%$, while PF-S-An2 attains the largest time efficiency, $e_{T,s}^{\text{msec}} = 46\%$. Overall, PLS-S-Z2 shows the strongest performance in this comparison, substantially outperforming the remaining JFS3dir1 and JFS3dir2 variants in terms of both residual evaluations and computational time.

stopping test: $f \leq 1\text{e-}05,$			$\text{sec} \leq 300,$			$\text{nf} \leq 20000$	
100 of 100 problems solved			# of anomalies			$e_{T,s}$ in %	
$\text{dim} \in [8, 135]$			#n #t #f			$e_{T,s}^{\text{nf}}$	$e_{T,s}^{\text{msec}}$
solver		solved	#n	#t	#f	$e_{T,s}^{\text{nf}}$	$e_{T,s}^{\text{msec}}$
PLS-S-Z2	Z2	100	0	0	0	89	87
PLS-S-Z1	Z1	100	0	0	0	44	36
PF-S-Z1	F-Z1	100	0	0	0	41	39
PF-S-Z2	F-Z2	100	0	0	0	39	38
PF-S-An1	F-An1	100	0	0	0	39	45
PF-S-An2	F-An2	100	0	0	0	39	46
PF-S-De	F-De	100	0	0	0	38	42
PLS-S-An2	An2	100	0	0	0	38	37
PLS-S-An1	An1	100	0	0	0	38	34
PLS-S-De	De	100	0	0	0	37	35

Table 4: Comparison between the JFS3dir1 and JFS3dir2 variants.

6 A comparison between the JFS3dir3 variants

Table 5 compares the JFS3dir3 variants in terms of robustness and mean efficiency. All variants solve the entire set of 100 problems without anomalies. In terms of residual evaluations, the fixed-step variants generally outperform their corresponding line-search variants. In particular, PF-S-A4 and PF-S-A1 attain the highest efficiency, with $e_{T,s}^{\text{nf}} = 77\%$, followed by PF-S-A5 and PF-S-A8 with 76%, and PF-S-A6 and PF-S-A7 with 75%. Among the line-search variants, PLS-S-A4, PLS-S-A1, PLS-S-A5, and PLS-S-A8 attain the highest residual-evaluation efficiency, with $e_{T,s}^{\text{nf}} = 55\%$. In terms of computational time, PF-S-A5 performs best overall, with $e_{T,s}^{\text{msec}} = 69\%$, followed

by PF-S-A4 and PF-S-A8 with 68%. Thus, the fixed-step realization is generally more efficient in terms of both residual evaluations and computational time across the corresponding JFS3dir3 variants.

stopping test: $f \leq 1\text{e-}05,$		$\text{sec} \leq 300,$			$\text{nf} \leq 20000$		
100 of 100 problems solved							
$\text{dim} \in [8, 135]$		# of anomalies			$e_{T,s}$ in %		
solver		solved	#n	#t	#f	$e_{T,s}^{\text{nf}}$	$e_{T,s}^{\text{msec}}$
PF-S-A4	F-A4	100	0	0	0	77	68
PF-S-A1	F-A1	100	0	0	0	77	66
PF-S-A5	F-A5	100	0	0	0	76	69
PF-S-A8	F-A8	100	0	0	0	76	68
PF-S-A6	F-A6	100	0	0	0	75	67
PF-S-A7	F-A7	100	0	0	0	75	67
PF-S-A3	F-A3	100	0	0	0	74	66
PF-S-A2	F-A2	100	0	0	0	74	63
PLS-S-A4	A4	100	0	0	0	55	48
PLS-S-A1	A1	100	0	0	0	55	40
PLS-S-A5	A5	100	0	0	0	55	48
PLS-S-A8	A8	100	0	0	0	55	46
PLS-S-A3	A3	100	0	0	0	52	46
PLS-S-A2	A2	100	0	0	0	52	42
PLS-S-A7	A7	100	0	0	0	51	40
PLS-S-A6	A6	100	0	0	0	51	40

Table 5: Comparison between the JFS3dir3 variants.

7 Efficiency Improvements of Fixed-Step over Line-Search Variants

From Tables 3–5, we conclude that the fixed-step PF-S realizations generally improve both the residual-evaluation efficiency and the computational-time efficiency relative to their line-search PLS-S counterparts. The main exception occurs for the JFS3dir1 directions, where PLS-S-Z2 substantially outperforms its fixed-step counterpart in both measures, while PLS-S-Z1 is slightly better in terms of residual evaluations. Figures 1–3 illustrate these differences directly by plotting the percentage-point improvements in $e_{T,s}^{\text{nf}}$ and $e_{T,s}^{\text{msec}}$.

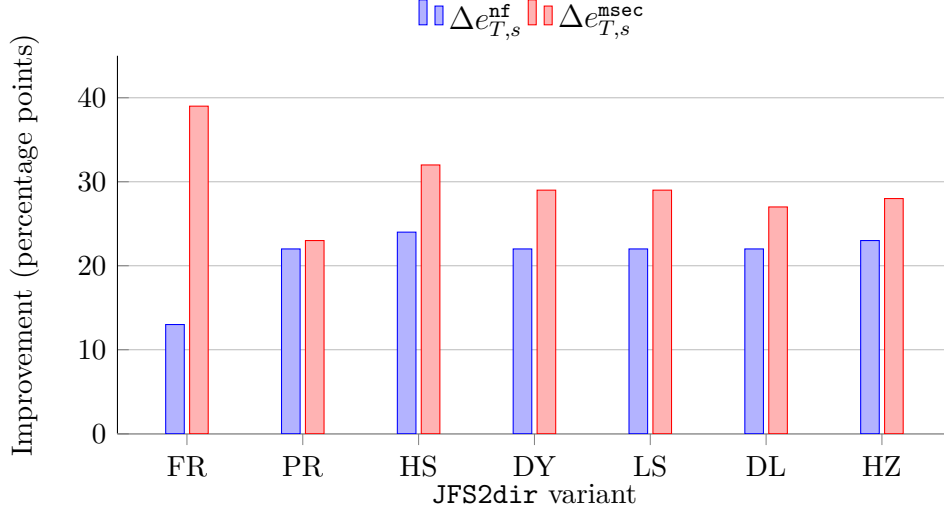


Figure 1: Efficiency improvement of the fixed-step PF-S variants over their line-search PLS-S counterparts for the JFS2dir family. Here, $\Delta e_{T,s} := e_{T,s}^{\text{PF-S}} - e_{T,s}^{\text{PLS-S}}$, so positive values favor the fixed-step variant and negative values favor the line-search variant.

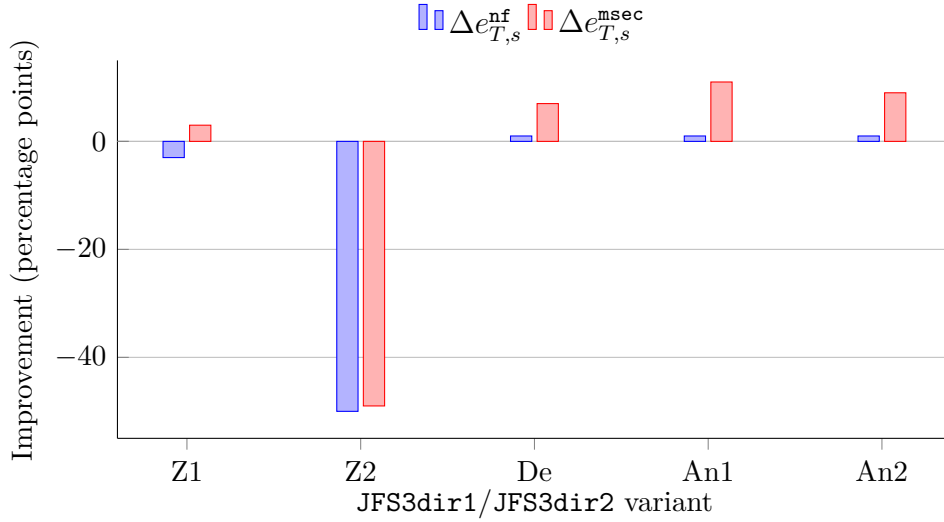


Figure 2: Efficiency improvement of the fixed-step PF-S variants over their line-search PLS-S counterparts for the JFS3dir1 and JFS3dir2 families. Positive values favor the fixed-step variant, whereas negative values favor the line-search variant.

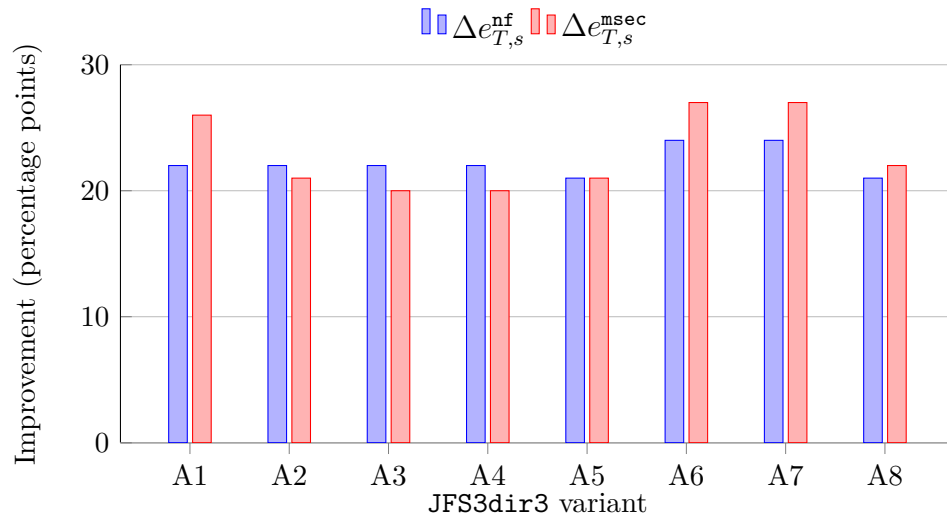


Figure 3: Efficiency improvement of the fixed-step PF-S variants over their line-search PLS-S counterparts for the JFS3dir3 family. Positive values favor the fixed-step variant, whereas negative values favor the line-search variant.

8 A comparison between OGDA-type variants

For the comparison methods, we use the unconstrained versions of EG, OGDA, EAG, and FOGDA. Their principal tuning parameter is the step size $s > 0$. For EG, the iterations are

$$w_k = x_k - sE(x_k), \quad x_{k+1} = x_k - sE(w_k),$$

whereas OGDA uses

$$x_{k+1} = x_k - 2sE(x_k) + sE(x_{k-1}),$$

after the initial step $x_1 = x_0 - sE(x_0)$.

For EAG, the implementation uses $\theta_k = 1/(k+2)$, and the iterations

$$w_k = x_k + \theta_k(x_0 - x_k) - sE(x_k), \quad x_{k+1} = x_k + \theta_k(x_0 - x_k) - sE(w_k).$$

Thus, the anchoring parameter is iteration dependent and is not tuned separately in the numerical experiments.

For FOGDA, the implementation uses the Fast-OGDA parameter $\alpha = 50 > 2$. With the auxiliary sequence denoted by w_k , the Fast-OGDA coefficients are determined from α , s , and the iteration index. The initialization used in the implementation is

$$w_0 = x_0, \quad x_1 = x_0 - sE(x_0).$$

Thereafter, for $k \geq 1$, the coefficient sequences are

$$\theta_k = 1 - \frac{\alpha}{k + \alpha}, \quad \eta_k = \frac{\alpha}{2(k + \alpha)}, \quad \kappa_k = \frac{1}{2} \left(1 + \frac{k}{k + \alpha} \right),$$

and the iterations are

$$w_k = x_k + \theta_k(x_k - x_{k-1}) - s\eta_k E(w_{k-1}), \quad x_{k+1} = w_k - s\kappa_k (E(w_k) - E(w_{k-1})).$$

No additional tuning of α is performed in the reported numerical experiments.

In the numerical experiments, the nominal step size supplied by the wrapper to all four methods is $s = 0.48/L$, where L is the Lipschitz constant associated with the corresponding test problem. The implementation additionally imposes the method-specific safety caps

$$s \leq 0.99/L \quad \text{for EG}, \quad s \leq 0.49/L \quad \text{for OGDA},$$

and

$$s \leq 0.25/L \quad \text{for } \mathbf{EAG}, \quad s \leq 0.10/L \quad \text{for } \mathbf{FOGDA}.$$

Consequently, the safeguards for **EG** and **OGDA** are not activated, whereas those for **EAG** and **FOGDA** reduce the nominal step size. The effective step sizes used in the experiments are therefore

$$s_{\mathbf{EG}} = s_{\mathbf{OGDA}} = 0.48/L, \quad s_{\mathbf{EAG}} = 0.25/L, \quad s_{\mathbf{FOGDA}} = 0.10/L.$$

Thus, all four comparison methods are supplied with the same nominal Lipschitz-scaled step size, while the implementation retains method-specific safety caps and uses the fixed choices $\theta_k = 1/(k+2)$ for **EAG** and $\alpha = 50$ for **FOGDA**.

Table 6 compares the **OGDA**-type variants in terms of robustness and mean efficiency. The **OGDA** and **EG** solvers are the most robust, solving all 100 problems without anomalies. Among them, **OGDA** is clearly the most efficient, attaining $e_{T,s}^{\text{nf}} = 88\%$ and $e_{T,s}^{\text{msec}} = 83\%$, compared with 44% and 34%, respectively, for **EG**. The **FOGDA** and **EAG** solvers are less robust, solving 79 and 65 problems, respectively. In terms of mean efficiency, **EAG** attains $e_{T,s}^{\text{nf}} = 32\%$ and $e_{T,s}^{\text{msec}} = 34\%$, whereas **FOGDA** attains 9% and 10%, respectively.

stopping test: $f \leq 1\text{e-}05,$			$\text{sec} \leq 300,$			$\text{nf} \leq 20000$	
100 of 100 problems solved							
$\text{dim} \in [8, 135]$			# of anomalies			$e_{T,s}$ in %	
solver		solved	#n	#t	#f	$e_{T,s}^{\text{nf}}$	$e_{T,s}^{\text{msec}}$
OGDA	OGDA	100	0	0	0	88	83
EG	EG	100	0	0	0	44	34
FOGDA	FOGDA	79	21	0	0	9	10
EAG	EAG	65	35	0	0	32	34

Table 6: Comparison between the **OGDA**-type variants.

References

- [1] A. Bohm, M. Sedlmayer, E. R. Csetnek, and R. I. Bot. Two steps at a time—taking gan training in stride with tseng’s method. *SIAM Journal on Mathematics of Data Science* 4 (2022), 750–771.

- [2] R. I. Boş, E. R. Csetnek, and D. K. Nguyen. Fast optimistic gradient descent ascent (OGDA) method in continuous and discrete time. *Foundations of Computational Mathematics* **25** (November 2023), 163–222.
- [3] T. Chavdarova, M. I. Jordan, and M. Zampetakis. Last-iterate convergence of saddle point optimizers via high-resolution differential equations. *arXiv preprint arXiv:2112.13826* (2021).
- [4] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio. Generative adversarial networks. *Communications of the ACM* **63** (2020), 139–144.
- [5] E. Gorbunov, H. Berard, G. Gidel, and N. Loizou. Stochastic extragradient: General analysis and improved rates. In *International Conference on Artificial Intelligence and Statistics*, pp. 7865–7901. PMLR (2022).
- [6] A. H. Ibrahim, P. Kumam, A. B. Abubakar, W. Jirakitpuwapat, and J. Abubakar. A hybrid conjugate gradient algorithm for constrained monotone equations with application in compressive sensing. *Heliyon* **6** (March 2020), e03466.
- [7] Y. Jiao and G. Li. Minimax-optimal multi-agent robust reinforcement learning. *arXiv preprint arXiv:2412.19873* (2024).
- [8] C. Jin, P. Netrapalli, R. Ge, S. M. Kakade, and M. I. Jordan. On nonconvex optimization for machine learning: Gradients, stochasticity, and saddle points. *Journal of the ACM* **68** (February 2021), 1–29.
- [9] M. Kimiaei, S. Shabani, and M. Breuß. Subspace methods for min-max problems (2026).
- [10] G. M. Korpelevich. The extragradient method for finding saddle points and other problems. *Matecon* **12** (1976), 747–756.
- [11] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu. Towards deep learning models resistant to adversarial attacks. (2018).
- [12] S. Omidshafiei, J. Pazis, C. Amato, J. P. How, and J. Vian. Deep decentralized multi-task multi-agent reinforcement learning under partial observability. In *International conference on machine learning*, pp. 2681–2690. PMLR (2017).
- [13] L. D. Popov. A modification of the Arrow-Hurwicz method for search of saddle points. *Mathematical notes of the Academy of Sciences of the USSR* **28** (1980), 845–848.

- [14] A. Robey, L. Chamon, G. J. Pappas, H. Hassani, and A. Ribeiro. Adversarial robustness with semi-infinite constrained learning. *Advances in Neural Information Processing Systems* **34** (2021), 6198–6215.
- [15] E. K. Ryu and S. Boyd. Primer on monotone operator methods. *Appl. comput. math* **15** (2016), 3–43.
- [16] A. Wood and B. Wollenberg. *Power Generation Operation and Control*. John Wiley and Sons (1996).
- [17] Y. Xiao and H. Zhu. A conjugate gradient method to solve convex constrained monotone equations with applications in compressive sensing. *J. Math. Anal. Appl* **405** (September 2013), 310–319.
- [18] T. Yoon and E. K. Ryu. Accelerated algorithms for smooth convex-concave minimax problems with $\mathcal{O}(1/k^2)$ rate on squared gradient norm. In *International Conference on Machine Learning*, pp. 12098–12109. PMLR (2021).